CS310 Natural Language Processing

Assignment 1 (written)

Total points: 10

Consider a softmax layer of output size k (k>1). Within unit *i* (*i*=1,…,k), the linear before activation is z_i, (also called logits), and the output after activation is o_i. We have the following equation for forward propagation:

$$o_i = \frac{\exp(z_i)}{\sum_{j=1}^{k} \exp(z_j)}$$

The cross-entropy loss between an output prediction $\hat{y} = [o_1, \dots, o_k]$ and the ground truth label $y = [0, \dots, 1 \dots, 0]$ (one-hot encoded) is defined as:

$$L_{CE} = -\sum_{i=1}^{k} y_i \log(\hat{y}_i) = -\sum_{i=1\dots k} y_i \log o_i$$

1) Show that when k=2, the loss is equivalent to binary cross-entropy (BCE) loss $L_{BCE} = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)$

(**4 points**, show your math derivation to get the credits)

2) What is the gradient w.r.t the *i*th unit $\frac{\partial L}{\partial z_i}$?

(**6 points**, show your math derivation to get the credits)